

Calibrating the alarm, not the interval: distribution-free notice guarantees for end-of-life alarms, with censored fleet data

ARTICLE INFO

Keywords:

A
B

ABSTRACT

Stub.

1. Introduction

A prognostic model creates value through the maintenance actions it triggers. For a component that is replaced once, near the end of its life, the action is usually triggered by an *end-of-life alarm*: a policy watches the unit cycle by cycle and raises one alarm, which schedules a replacement after a lead time for parts, crew and access. The first reliability requirement on such an alarm is therefore not the accuracy of a remaining useful life (RUL) estimate at some cycle chosen by the evaluator, but the probability that the unit fails before the replacement the alarm scheduled. This paper gives that probability a distribution-free, finite-sample guarantee.

The usual ways of evaluating and calibrating prognostics do not control it. Accuracy metrics and asymmetric acceptance windows score an estimate at a time, and a window score at the alarm mixes the estimate with the alarm time [1, 2]. Decision-oriented metrics, such as the metric M of Kamariotis et al. [3], score and tune policies by long-run cost but give no guarantee for a new unit. Conformal prediction gives RUL intervals with distribution-free coverage [4–6], but the coverage holds at a cycle chosen independently of the data, whereas an alarm acts at the first cycle at which the estimate, or its lower bound, crosses a level: a stopping time at which the estimate is, by selection, at its most optimistic. Two further obstacles stand between any calibration and the field. Fleet data are censored, because most units are replaced before they fail; and populations change, from one production batch or operating campaign to the next.

This paper calibrates the alarm itself. For a threshold trigger on any causal signal, the notice left by the alarm is monotone in the warning level, so each calibration unit has a single *critical level*: the lowest warning level at which its alarm would still have left more than the lead time. The split-conformal quantile of these critical levels is a warning level with a finite-sample, distribution-free guarantee that a new exchangeable unit is replaced before it fails with probability at least $1 - \alpha$. The contributions are:

1. **Theory.** A closed-form critical level for debounced threshold triggers (Lemma 2); the notice guarantee with a matching upper bound (Theorem 1); a decision-aware choice of the level that keeps the

guarantee (Corollary 1); a cost bound and a certification floor that state what n calibration units can certify (Proposition 1); a pathwise censoring bound under which the guarantee holds for *any* censoring mechanism, so that units replaced before failure can be used for calibration (Lemma 3, Theorem 2); and a cross-fitted variant that calibrates on all training units (Theorem 3). A supporting result (Proposition 2) explains why window scores of threshold alarms measure timing and cannot certify notice.

2. **Method.** The conformal notice alarm (CNA) and its decision-aware and cross-fitted variants (Algorithm 1), which wrap any signal, including a learned regressor and a physically motivated capacity trend.

3. **Evidence.** Two protocols hashed before their experiments; seven datasets (PHM08, C-MAPSS FD001–FD004, N-CMAPSS and 123 lithium-ion cells in three batches); comparisons with interval-calibrated alarms, window- and cost-tuned alarms, the probability-threshold policy of Kamariotis et al. (also in their own decision setting and metric), age replacement and a capacity-trend baseline; three censoring mechanisms; and transfer to held-out battery batches and N-CMAPSS subsets.

Section 2 reviews related work, Section 3 states the problem, Section 4 gives the theory and Section 5 the method. Section 6 describes the design, Section 7 the results and Section 8 their implications and limits.

2. Related work

Evaluating prognostics through decisions. Remaining useful life (RUL) estimates are usually scored at times chosen by the evaluator, with accuracy metrics or asymmetric scores that penalise late predictions [1, 2, 7]. A line of work in this journal evaluates prognostics through the maintenance decisions they trigger. de Pater et al. [8] schedule aircraft maintenance from alarms raised on RUL prognostics and protect against prognostic error with a safety factor; Kamariotis et al. [3] propose a metric M , the relative excess long-run cost of a policy over perfect prognostics, and tune probability-threshold policies with it; dynamic

predictive-maintenance policies act on predicted failure probabilities [9, 10]. Age replacement is the benchmark that any condition-based policy must beat [11, 12]. None of these gives a finite-sample guarantee on the notice an alarm leaves.

Conformal prediction for RUL. Conformal prediction turns any point predictor into prediction sets with finite-sample, distribution-free coverage under exchangeability [13–15]. Javanmardi and Hüllermeier [4] conformalised gradient boosting and convolutional networks to obtain RUL intervals on C-MAPSS; Robinson [5] used asymmetric conformalised quantile regression [16] to make the lower RUL bound conservative; Wang et al. [6] combined conformal prediction with imputation of missing sensor data and noted that exchangeability is a concern for degradation data. In all of these the calibrated object is the interval at a given cycle. Candès et al. [17] give conformal lower bounds on survival times under censoring, at a fixed covariate value and under assumptions on the censoring mechanism. An alarm, however, is a stopping rule, evaluated at the cycle at which the bound first crosses a level; per-cycle coverage does not imply coverage at that time, and Section 7.1 shows that the difference is large.

Risk control, cross-fitting and shift. Conformal risk control [18] calibrates a scalar parameter so that the expected value of a monotone loss is bounded, and nested conformal prediction [19] expresses conformal procedures through nested families of sets. Our construction is an instance of the nested view, with the warning level as the nesting parameter and failure before the planned replacement as the loss. What is specific is the closed-form critical level of a debounced trigger (Lemma 2), the pathwise censoring bound that makes calibration possible with replaced units (Theorem 2), the decision-aware choice that keeps the guarantee (Corollary 1), and the certification floor (Proposition 1). Cross-fitting follows the jackknife+/CV+ construction of Barber et al. [20]. When calibration and new units are not exchangeable, weighted and non-exchangeable conformal methods bound the coverage gap [21, 22]; they need a model of the shift, which we do not assume. We instead measure the failure of exchangeability across battery batches and N-CMAPSS subsets and the cost of restoring it with pilot units.

Relation to our earlier evaluation. This article supersedes an evaluation study of the same alarm recipe on the same data [23], which separated prediction accuracy, alarm timing, maintenance cost and transfer and proposed no method. Its numbers are cited as prior results; every number presented as new was produced by the code released with this article.

3. Problem formulation

Units, alarms and notice. A unit observed at rows $j = 1, \dots, n_u$ (cycles $t_1 < \dots < t_{n_u}$) has life L and remaining life $r_j = L - t_j \geq 1$, decreasing in j . A causal signal s_j , here a smoothed RUL estimate, is computed from rows $1, \dots, j$. A *threshold trigger* with debounce k and warning level H raises an alarm at the first row at which the signal has been at or below H for k consecutive rows. Writing $m_j = \max(s_{j-k+1}, \dots, s_j)$ for $j \geq k$ ($m_j = +\infty$ otherwise) and $\rho_j = \min_{i \leq j} m_i$ for the *debounced running minimum*, the alarm row is

$$\tau(H) = \min\{j : \rho_j \leq H\} \quad (\tau = \infty \text{ if there is no such row}),$$

and the *notice* is $N(H) = r_{\tau(H)}$ ($N = 0$ without an alarm).

Maintenance consequence. An alarm schedules a replacement ℓ cycles later (the *lead time*). If $N(H) > \ell$ the planned replacement happens at $\tau(H) + \ell$ at cost 1; otherwise, or without an alarm, the unit fails at L at cost $\rho > 1$. Units are renewed, so by the renewal-reward theorem the long-run cost per cycle is $E[\text{cost}]/E[\text{use}]$, estimated by total cost over total cycles of use and reported relative to perfect foresight (replacement at $L - 1$, cost 1) [11]. The reliability requirement we target is

$$P(N(H) \leq \ell) \leq \alpha,$$

the probability that the unit fails before the replacement its alarm scheduled.

Window score. The earlier study scored the first alarm by an asymmetric window: with reported estimate $\hat{Y}$ and error $e = \hat{Y} - R$, $R = N(H)$, the alarm is correct if $a \leq e \leq b$, with $[a, b] = [-13, +2]$ cycles for the engines and $[-39, +6]$ cycles for the battery cells.

4. Theory

4.1. The critical level and the notice guarantee

Two elementary properties of the trigger make the notice calibratable.

Lemma 1 (Monotonicity). For $H \leq H'$, $\tau(H') \leq \tau(H)$ and $N(H') \geq N(H)$.

Proof. ρ is non-increasing, so $\{j : \rho_j \leq H\} \subseteq \{j : \rho_j \leq H'\}$; the first element of the larger set is not later, and r_j decreases in j . $\square$

Lemma 2 (Critical level). Let $j_\ell = \max\{j : r_j > \ell\}$ and define the critical level $V(\ell) = \rho_{j_\ell}$ ($V = +\infty$ if no row has $r_j > \ell$). Then for every finite H , $N(H) > \ell \iff H \geq V(\ell)$.

Proof. Because r decreases in j , $N(H) > \ell$ iff the alarm is raised at a row $\tau(H) \leq j_\ell$, iff $\rho_j \leq H$ for some $j \leq j_\ell$, iff $\rho_{j_\ell} \leq H$, because ρ is non-increasing. $\square$

V is the lowest warning level at which the unit's alarm would still have left more than ℓ cycles. It is one number per unit, computed from its signal and its life in a single pass, and it is all that calibration needs.

Theorem 1 (Notice guarantee). *Fix the signal map (for example a regressor fitted on units disjoint from those below), the smoothing, the debounce k and the lead time ℓ , and assume that every unit has at least k rows with $r_j > \ell$, so that $V < \infty$. Let the calibration units $1, \dots, n$ and a new unit $n+1$ be exchangeable, with critical levels $V_1, \dots, V_{n+1}$. For $\alpha \in [1/(n+1), 1)$ let $q = \lceil (n+1)(1-\alpha) \rceil$ and $\hat{H} = V_{(q)}$, the q -th smallest of $V_1, \dots, V_n$. Then*

$$P(N_{n+1}(\hat{H}) > \ell) \geq 1 - \alpha, \quad (1)$$

where the probability is over the calibration units and the new unit. If the V_i are almost surely distinct, the probability is also at most $1 - \alpha + 1/(n+1)$.

Proof. By Lemma 2 the event is $\{V_{n+1} \leq \hat{H}\}$. If $V_{n+1} > V_{(q)}$, at least q calibration values are strictly smaller than V_{n+1} , so the rank of V_{n+1} among all $n+1$ values (ties broken uniformly at random) is at least $q+1$. By exchangeability the rank is uniform on $\{1, \dots, n+1\}$, so $P(V_{n+1} > \hat{H}) \leq (n+1-q)/(n+1) \leq \alpha$. With distinct values, $V_{n+1} \leq V_{(q)}$ iff the rank is at most q , which has probability $q/(n+1) < 1 - \alpha + 1/(n+1)$. $\square$

For $\alpha < 1/(n+1)$ the level is not attainable with n calibration units, and the procedure reports this instead of a level.

Corollary 1 (A decision-aware choice keeps the guarantee). *Let $q_0 = \lceil (n+1)(1-\alpha_{\max}) \rceil$ and let $\hat{q} \geq q_0$ be any index chosen from the calibration data, for example to minimise an estimated cost. Then $P(N_{n+1}(V_{(\hat{q})}) > \ell) \geq 1 - \alpha_{\max}$.*

Proof. $V_{(\hat{q})} \geq V_{(q_0)}$; by Lemma 1 the event for $V_{(\hat{q})}$ contains the event for $V_{(q_0)}$, whose probability is at least $1 - \alpha_{\max}$ by Theorem 1. $\square$

Proposition 1 (Cost bound and certification floor). (i) *Under the conditions of Theorem 1, the expected cost of one renewal cycle of a new unit is at most $1 + (\rho-1)\alpha$, and its long-run cost rate is at most $(1 + (\rho-1)\alpha)/E[\text{use}]$.* (ii) *If the critical levels are almost surely distinct, the alarm at $V_{(q)}$ fails before replacement with probability exactly $(n+1-q)/(n+1)$. Hence no level chosen among the order statistics has a failure probability below $1/(n+1)$, the expected cost of a renewal cycle is then at least $1 + (\rho-1)/(n+1)$, and certifying a failure probability α requires $n \geq 1/\alpha - 1$ calibration units.*

Proof. (i) The cost of a cycle is $1 + (\rho-1)\mathbf{1}\{N \leq \ell\}$; take expectations and apply the renewal-reward theorem. (ii) With distinct values the rank of V_{n+1} is uniform, and the alarm fails iff the rank exceeds q , with probability $(n+1-q)/(n+1)$, which is smallest at $q = n$. $\square$

Proposition 1(ii) is the price of a distribution-free certificate. With 32 calibration units and a failure that costs as much as fifty planned replacements, any order-statistic level carries an expected excess of at least $49/33 \approx 1.5$ replacement costs per renewal over a policy that never fails. Rules that choose levels beyond the most demanding calibration unit can do better on a given test set, but nothing certifies them.

4.2. Censored calibration data

In a fleet most units are replaced before they fail, so their lives, and hence their critical levels, are not observed. A replaced unit is still informative.

Lemma 3 (Censoring gives an upper bound). *If a unit is known to be alive at cycle c ($L > c$), then $V(\ell) \leq U(c)$, where $U(c) = \rho_{j_c}$ with $j_c = \max\{j : t_j \leq c - \ell\}$ ($U = +\infty$ if no such row). $U(c)$ is the critical level the unit would have had if it had failed at c .*

Proof. For rows with $t_j \leq c - \ell$, $r_j = L - t_j > c - t_j \geq \ell$, so $j_\ell \geq j_c$ and $V = \rho_{j_\ell} \leq \rho_{j_c}$ because ρ is non-increasing. If the unit had failed at c , its last row with more than ℓ cycles left would be j_c . $\square$

Theorem 2 (Guarantee under arbitrary censoring). *Under the conditions of Theorem 1, let each calibration unit report $W_i = V_i$ if its failure was observed and $W_i = U_i(c_i)$ if it was last seen alive at c_i , where the censoring times may depend in any way on the units, their signals and each other. Then $\hat{H}_W = W_{(q)}$ satisfies $P(N_{n+1}(\hat{H}_W) > \ell) \geq 1 - \alpha$.*

Proof. By Lemma 3, $W_i \geq V_i$ for every i , so every order statistic satisfies $W_{(q)} \geq V_{(q)}$. By Lemma 1 the event for $\hat{H}_W$ contains the event for $V_{(q)}$, whose probability is at least $1 - \alpha$ by Theorem 1. No assumption on the censoring mechanism is used, because the comparison is pathwise. $\square$

In words: treat every censored unit as if it had failed when it was last seen alive. The guarantee survives any censoring mechanism, including informative censoring, at the price of conservatism that grows with how early units are censored. Dropping censored units (complete-case analysis) keeps the guarantee only when censoring is independent of the unit. A useful special case is censoring by an earlier alarm policy: a unit replaced ℓ cycles after an alarm at level H_0 has $U \leq H_0$, so if H_0 lies below the target quantile, such units do not move $\hat{H}_W$ at all (Section ??).

4.3. Calibration on all training units by cross-fitting

The split design spends calibration units that could have trained the signal. Cross-fitting uses every training unit for both, at the price of a weaker guarantee. Partition the n training units into K folds, let μ_{-k} be

the signal model fitted without fold k by a symmetric algorithm, let V_i be the critical level of unit i under $\mu_{-k(i)}$, and let $\rho_t^{(k)}$ be the debounced running minimum of a new unit under μ_{-k} . The cross-fitted alarm (CNA+) is raised at the first row at which

$$C_t = \#\{i : \rho_t^{(k(i))} \leq V_i\} \geq n+1 - \lceil (1-\alpha)(n+1) \rceil. \quad (2)$$

C_t is non-decreasing in t , so the rule is a trigger.

Theorem 3 (Cross-fitted guarantee). *If the n training units and the new unit are exchangeable and the folds have equal size n/K , then*

$$P(N_{n+1} > \ell) \geq 1 - 2\alpha - \min\left\{\frac{2(1-1/K)}{n/K+1}, \frac{1-K/n}{K+1}\right\}.$$

Proof. Let $V_{n+1}^{(k)}$ be the critical level of the new unit under μ_{-k} . By Lemma 2 applied under each model, $\rho_{j_\ell}^{(k)} \leq V_i \iff V_{n+1}^{(k)} \leq V_i$, so the alarm fails to leave more than ℓ cycles iff $C_{j_\ell} < n+1 - q$, that is iff $\#\{i : V_{n+1}^{(k(i))} > V_i\} \geq q$ with $q = \lceil (1-\alpha)(n+1) \rceil$. This is the non-coverage event of the CV+ interval of Barber et al. [20] with the nonconformity score “critical level of a unit under a model not fitted on it”. Their proof of the CV+ bound (their Theorem 4) compares scores through a tournament matrix built from leave-fold-out models and uses only the exchangeability of the units and the symmetry of the fitting algorithm, not the form of the score, so it applies verbatim and gives the stated bound. $\square$

The bound is loose for small n (with $n = 80$ and $K = 5$ the finite- K term is about 0.09), and, as for CV+ intervals [20], realised failure rates are usually close to α ; Section ?? reports them.

4.4. What a window score and a per-cycle interval certify

Proposition 2 (Window score of an estimate versus a constant). *Let $R \in \mathbb{Z}$ be the notice at the alarm, $\hat{Y}$ the reported estimate, $[a, b]$ the acceptance window for $e = \hat{Y} - R$, c any constant and $D = \hat{Y} - c$. With the half-open intervals $I_b = [c - b, c - b + D)$ and $I_a = (c - a, c - a + D]$ for $D \geq 0$ (mirrored for $D < 0$): (i) $|\mathbf{1}\{a \leq \hat{Y} - R \leq b\} - \mathbf{1}\{a \leq c - R \leq b\}| \leq \mathbf{1}\{R \in I_a\} + \mathbf{1}\{R \in I_b\}$; (ii) if, given $\hat{Y}$, the probability mass function of R is at most π_a within Δ of $c - a$ and at most π_b within Δ of $c - b$, and $|D| \leq \Delta$, then the window scores satisfy $|S_{\text{est}} - S_c| \leq (\pi_a + \pi_b) E[|D|]$; (iii) if the reported estimate is the triggering signal, $\hat{Y} = s_\tau$, and $\tau > k$, then $H - k\delta^+ \leq \hat{Y} \leq H$ with $\delta^+ = \max(0, \max_{\tau-k < i \leq \tau} (s_{i-1} - s_i))$.*

Proof. (i) For $D \geq 0$ the estimate accepts $R \in [c - b + D, c - a + D]$ and the constant $R \in [c - b, c - a]$. R accepted only by the estimate satisfies $c - a < R \leq c - a + D$; accepted only by the constant, $c - b \leq R < c - b + D$. The case $D < 0$ is symmetric, and the

argument holds also when the two sets are disjoint. (ii) Take expectations of (i); a half-open interval of length $|D|$ contains at most $\lceil |D| \rceil$ integers. (iii) $\hat{Y} = s_\tau \leq m_\tau \leq H$. As the alarm was not raised at $\tau - 1 \geq k$, some row $i_0 \in [\tau - k, \tau - 1]$ has $s_{i_0} > H$, and $s_\tau = s_{i_0} - \sum_{i=i_0+1}^\tau (s_{i-1} - s_i) > H - k\delta^+$. $\square$

For a threshold-crossing alarm the reported estimate is pinned to a band of width $k\delta^+$ below the level, so a window score is, up to the edge term of (ii), the probability that the notice falls in a band $[c - b, c - a]$. It can be high while the notice is too short for the lead time, so it cannot certify the event of Theorem 1. Nor can a per-cycle conformal lower bound $\text{LB}_t = s_t - q_\alpha$: its coverage $P(r_t \geq \text{LB}_t) \geq 1 - \alpha$ holds for a row drawn independently of the data, whereas the alarm “act when $\text{LB}_t \leq \ell + 1$ ” is read at the first row at which the bound crosses a level, where an optimistic error is most likely. That alarm is a threshold trigger at level $\ell + 1 + q_\alpha$ on the same signal, so it differs from CNA only in how the level is calibrated, which makes it the natural comparator (Section 7.1).

5. Method: the conformal notice alarm

Algorithm 1 Conformal notice alarm (CNA) and its variants

Input: causal signal map; debounce k ; lead time ℓ ; level α ; calibration units disjoint from those used to fit the signal.

Calibrate. For each calibration unit i : compute its debounced running minimum $\rho^{(i)}$ and its critical level $V_i = \rho_{j_\ell}^{(i)}$ (Lemma 2). If the unit was replaced or last seen alive at cycle c_i , use $U_i(c_i)$ instead (Lemma 3). Set $\hat{H} = V_{(\lceil (n+1)(1-\alpha) \rceil)}$.

Operate. Raise the alarm at the first row at which the signal has been at or below $\hat{H}$ for k consecutive rows; schedule the replacement ℓ cycles later.

CNA-DA. Given a cost ratio ρ and a ceiling $\alpha_{\max}$, use $V_{(\hat{q})}$ with $\hat{q} = \arg \min_{q \geq q_0} [1 + (\rho - 1) \frac{n+1-q}{n+1}] / \text{use}(V_{(q)})$, $q_0 = \lceil (n+1)(1 - \alpha_{\max}) \rceil$, with the mean cycles of use estimated on the calibration units.

CNA+. Replace the split by K -fold cross-fitting and raise the alarm by rule (2).

Calibration costs one pass over the calibration series (Fig. 1). The numerator of the CNA-DA objective replaces the empirical failure rate by its conformal value, which is never zero: with $q = n$ it is $1/(n+1)$. This is the difference from plain empirical cost minimisation, which sees no failure among calibration units at the edge of the feasible region and was fragile in the earlier study [23]. By Corollary 1 the failure probability of CNA-DA is at most $\alpha_{\max}$ whatever ρ is.

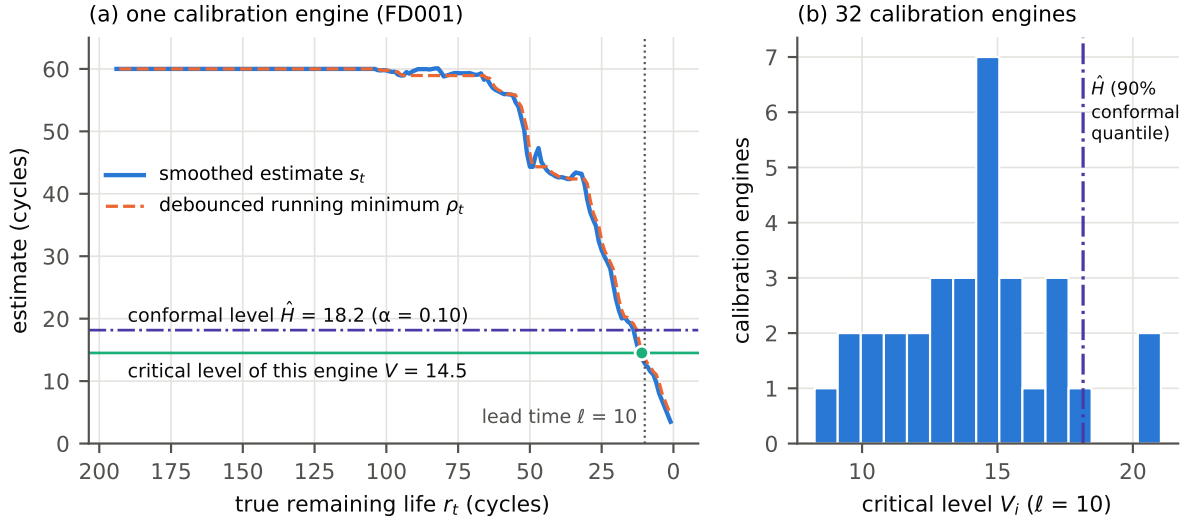


Fig. 1: The conformal notice alarm on FD001 (fold 1, lead time 10 cycles). (a) The smoothed estimate of one calibration engine and its debounced running minimum ρ_t ; the critical level V is the value of ρ at the last cycle with more than ℓ cycles left (dot). (b) Critical levels of the 32 calibration engines and the conformal level $\hat{H}$ at $\alpha = 0.10$.

6. Experimental design

The protocol (hypotheses H1–H5, datasets, policies, metrics and decision rules) was written and hashed with SHA-256 before the main experiments, and an amendment (H6–H8: censoring, cross-fitting and the setting of Kamariotis et al.) was hashed before its experiments, after the main results were known. The hashes are not registry preregistrations; they show that the files have not changed since, and their timing rests on the repository history. The code was smoke-tested on one fold before each hash.

Data. We use the 218 PHM08 training engines (the development dataset of the earlier study), the C-MAPSS FD001–FD004 training sets (100, 260, 100 and 249 engines) [1], N-CMAPSS reduced to one row per flight (99 engines in nine subsets, six temperature and speed channels) [24], and 123 lithium-ion cells of Severson et al. [25] in three batches (36, 43 and 44 cells; end of life at 80% of nominal capacity), aggregated into blocks of 3 cycles fixed a priori. The engine datasets are simulations from one simulator family; the battery data are laboratory measurements. The 17 PRONOSTIA bearings [26] of the earlier study are excluded: with at most about six calibration bearings per split, $\alpha \leq 0.10$ is not attainable.

Splits and information. Outer splits are five-fold cross-validation by unit (seed 2026), in which test units are exchangeable with training units, and, for transfer, leave-one-batch-out on the battery cells and leave-one-subset-out on N-CMAPSS. Inside each outer training set, 60% of the units (proper training) fit the normalisation and the regressors and 40% (calibration) are used for every selection, tuning and calibration step of every policy; CNA+ instead cross-fits all training

units. Outer test units are scored once. The proper-training/calibration split is repeated three times (repetition 0 is primary). Disjointness of the unit sets is asserted in the code; features at row t use rows $\leq t$ only (tested by truncation); no test life enters any setting.

Policies. The signal is the regressor of the earlier study: histogram gradient boosting with absolute-error loss (500 iterations, learning rate 0.05, 31 leaves, at least 40 samples per leaf) on causal features of regime-normalised sensors (early-life baseline, exponentially weighted means, local slopes, age), target $\min(r, c)$ [27]. Table 1 lists the policies. CNA uses settings fixed a priori (cap 60 cycles, battery 180; span 3; debounce 2), so that no selection touches the calibration units before the conformal step. Costs use lead times $\ell \in \{0, 5, 10, 15\}$ cycles (battery $\{0, 15, 30, 45\}$) and cost ratios $\rho \in \{5, 10, 20, 50\}$; the primary lead time is 10 cycles (battery 30). The cost ratios are illustrative.

Metrics, tests and decision rules. The primary metric is the *failure-before-replacement rate* on test units, with Clopper–Pearson intervals [28] and a one-sided exact binomial test of rate $> \alpha$, Holm-adjusted over datasets [29]. Notice at the alarm and wasted life measure what the guarantee costs. Cost rates are compared by paired bootstrap ratios over test units (2,000 resamples); a policy is called cheaper than another on a dataset if it is significantly cheaper in at least 8 of the 16 cost cells and significantly dearer in none. Window scores of estimate and constant are compared by the Tango score interval for paired proportions [30] and two one-sided tests [31] with a margin of 5 points. Intervals cover the sampling of test units given fitted policies; the repetitions show the variability from the split.

Table 1

Policies compared. All settings of learned policies are selected on calibration units; regressors are fitted on proper-training units (CNA+: cross-fitted on all training units).

Policy	Signal and trigger	Selected on calibration units	Guarantee
CNA ($\alpha = 0.05, 0.10, 0.20$)	GBM estimate (cap 60; battery 180 cycles), span 3, debounce 2, fixed a priori	$\hat{H}$ = conformal quantile of critical levels	$P(\text{fail}) \leq \alpha$ (Thm 1)
CNA-DA	As CNA	Order statistic minimising the cost bound, $\alpha_{\max} = 0.10$	≤ 0.10 (Cor. 1)
CNA+ / CNA+-DA	As CNA, five cross-fitted models	Count threshold of rule (2)	$\leq 2\alpha + O(K/n)$ (Thm 3)
LB-all / LB-near	As CNA; alarm when $s_t - q_\alpha \leq \ell + 1$	Conformal quantile of per-row residuals (all rows / rows with $r \leq \text{cap}$)	Per-row only
STW, window-tuned	GBM estimate (cap 40/60/90), span, debounce, shift	Most correct first warnings per H ; H by calibration cost	None
STW, cost-tuned (plain, 1-SE)	GBM estimate, span, debounce, threshold (2,160 settings)	Lowest calibration cost; 1-SE: largest notice within one bootstrap SE	None
Kamariotis P1 (1/ ρ , optimised)	$P(r \leq \ell + \Delta t \mid \text{estimate bin})$ from calibration rows (P-res)	Threshold 1/ ρ , or optimised on calibration cost	None
Age replacement	Age	Replacement age on training lives	None
Capacity trend (battery)	Discharge capacity extrapolated to 0.88 Ah	As the STW and CNA counterparts, on all training cells	CNA version: Thm 1

7. Results

All numbers are on outer test units. Calibration sets had 32 units (FD001, FD003, N-CMAPSS), 39 (battery), 70 (PHM08), 80 (FD004) and 83 (FD002) under cross-validation. Unless stated, rates are averaged over the three repetitions of the proper-training/calibration split, and tests use repetition 0, in which every unit is tested once.

7.1. The guarantee holds; interval calibration does not deliver it

Table 2 and Fig. 2 give the failure-before-replacement rate at the primary lead time. With $\alpha = 0.10$ the conformal notice alarm failed on 6.0–11.7% of test units across the seven datasets, and with $\alpha = 0.05$ and 0.20 on 1.3–5.1% and 14.3–20.6%. Pooled over the seven datasets, four lead times and three repetitions (13,788 unit–cell evaluations), the rates were 3.8, 9.3 and 19.4% for nominal 5, 10 and 20%. In repetition 0 no exact binomial test of rate > 0.10 was significant after Holm’s correction (smallest adjusted $p = 0.84$), so H1 is supported on all seven datasets. Rates slightly above 10% (FD003 11.7%, N-CMAPSS 11.1%, battery 10.6%, FD004 10.4%) are within sampling error; for distinct critical levels the expected rate is $(n+1-q)/(n+1)$, 9.1% with 32 calibration units and 10.0% with 39.

The interval-calibrated alarm on the same signal and trigger did not deliver its nominal level. Calibrated with residuals over all rows (LB-all), the lower-bound alarm at nominal 10% failed on 18.2% (N-CMAPSS) to 69.6% (battery) of units, 46.6% pooled, with $p < 0.005$ on every dataset (H2 supported, 7 of 7). Rows far from failure, where the capped target makes residuals small, dominate the pooled residual quantile, so the bound is tight far from failure and loose at the stopping

time. Restricting the residuals to rows with $r_t \leq c$ (LB-near) brought the rate close to nominal on six datasets (4.0–8.6%) but not on N-CMAPSS (15.2%, and 17.8% at the longest lead time), and its level depends on how the evaluator chooses the rows. The guarantee costs notice: at $\alpha = 0.10$ CNA alarms left a median of 15–17 engine cycles at a lead time of 10, against 11–12 for LB-all; on the battery cells 43 cycles at a lead time of 30.

7.2. Transfer: the guarantee does not survive a new batch unless it is recalibrated

With a whole battery batch held out, calibration and test units are no longer exchangeable, and the guarantee failed as the protocol predicted (H5a). Calibrated on the other two batches, CNA at $\alpha = 0.10$ failed on 0% of batch 1 but on 74% of batch 2 and 61% of batch 3 (Table 3, Fig. 3). The shortest-lived batch was the worst, the direction expected when a regressor trained mostly on longer-lived cells overestimates remaining life. CNA-DA failed on 45% and 36%, and in the worst cost cell it cost 24 times as much as the cheapest policy. The same loss of validity appeared on N-CMAPSS with a flight-data subset held out: 29–40% failures on three of nine subsets and 0–2% on the other six. Conservative uncertified rules did better under this shift: the one-standard-error cost-tuned alarm failed on 0–2% of the held-out battery cells and the capacity-trend policies on none. Their extra margin, which cost them little within batches, protected them across batches; a calibrated alarm has, by design, no margin beyond what its calibration units require.

Recalibration with pilot units of the new batch restored the guarantee (H5b). Using 10 randomly chosen cells of the held-out batch as calibration units and testing on the rest, the mean failure rate over 300

Table 2

Failure before the planned replacement at the primary lead time (10 cycles; battery 30 cycles), five-fold cross-validation, mean of three repetitions. Second column: repetition 0 for CNA at $\alpha = 0.10$ (failures; rate; Clopper–Pearson 95 % interval). Last column: median notice at the alarm (cycles), CNA 0.10 / LB-all 0.10.

Dataset (test units)	CNA 0.10, rep. 0	CNA 0.05	CNA 0.10	CNA 0.20	CNA-DA	LB-all 0.10	LB-near 0.10	Notice
PHM08 (218)	18 (0.083; 0.050–0.127)	0.034	0.086	0.177	0.009	0.446	0.080	16 / 11
FD001 (100)	7 (0.070; 0.029–0.139)	0.013	0.060	0.143	0.013	0.333	0.043	17 / 12
FD002 (260)	20 (0.077; 0.048–0.116)	0.042	0.088	0.191	0.008	0.381	0.040	15 / 12
FD003 (100)	14 (0.140; 0.079–0.224)	0.027	0.117	0.197	0.027	0.480	0.063	16 / 11
FD004 (249)	31 (0.124; 0.086–0.172)	0.051	0.104	0.206	0.011	0.466	0.086	16 / 11
N-CMAPSS (99)	11 (0.111; 0.057–0.190)	0.037	0.111	0.178	0.057	0.182	0.152	23 / 19
Battery (123)	15 (0.122; 0.070–0.193)	0.033	0.106	0.198	0.011	0.696	0.070	43 / 28

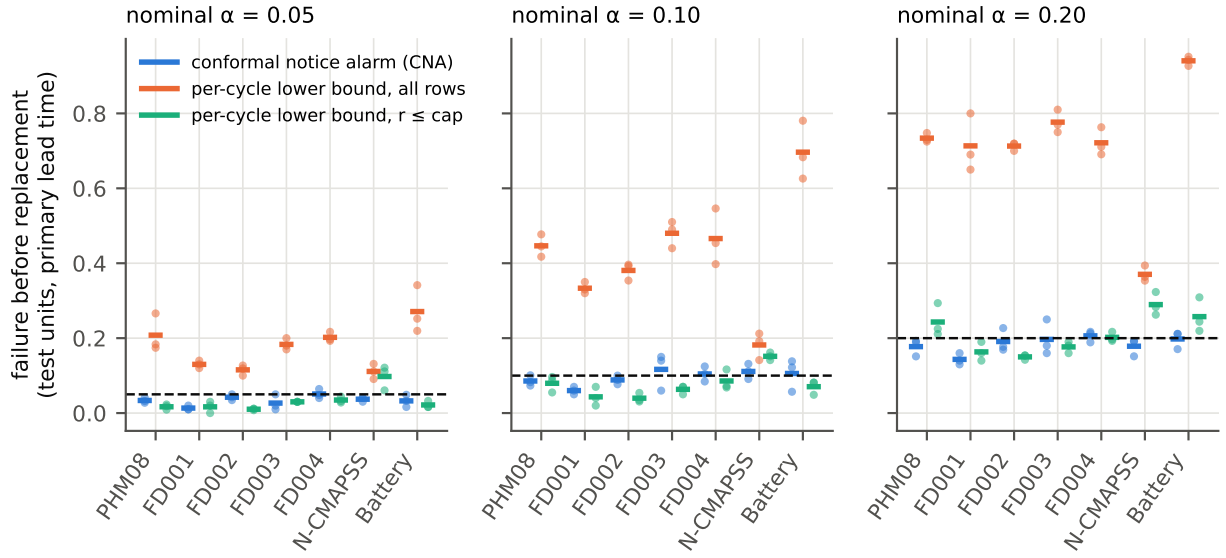


Fig. 2: Failure before the planned replacement at the primary lead time (10 cycles; battery 30) for the conformal notice alarm (CNA) and per-cycle split-conformal lower-bound alarms on the same signal and trigger, at three nominal levels (dashed). Dots: the three repetitions; bars: their mean. Five-fold cross-validation.

draws was 7.5–10.1 % at $\alpha = 0.10$ across batches, lead times and repetitions (8.4–9.5 % at the primary lead time), around the exact value $1/11 = 9.1\%$ of Proposition 1; with 20 pilot cells it was 8.3–10.4 %. Mean notice at a planned replacement was 48–60 cycles with 10 pilot cells, against 37 cycles on batches 2 and 3 under cross-batch calibration, where most cells failed. Five pilot cells cannot certify $\alpha = 0.10$ ($1/(m+1) > \alpha$) and certify $\alpha = 0.20$ with a failure rate of 16.5–16.7 %. The capacity-trend signal behaved the same way: calibrated across batches it failed on 56 % of batch 2, and with 10 pilot cells on 9 %. Because pilot cells need not run to failure (Theorem 2), a new batch can be calibrated from the first cells replaced under a conservative interim level; Section ?? shows that censoring of this kind costs little.

7.3. What the window score measured

Proposition 2 says that the window scores of the estimate and of a constant can differ only through units

whose notice falls near an edge of the constant's acceptance band. On every dataset the reported estimate was pinned below the level as part (iii) states (100 % of alarms within $[H - k\delta^+, H]$) and varied much less than the notice: its standard deviation was 2.3–3.2 cycles on the engines against 4.9–18.0 for the notice (battery 6.3 against 12.9). Discordant units, correct under one report and not the other, were 6.9–14.0 % on PHM08 and C-MAPSS, 11.4 % on the battery cells and 30.3 % on N-CMAPSS (Table 4).

The equivalence test of the protocol was not supported. With a margin of ± 5 points, estimate and constant were equivalent only on FD002; on FD001, FD003 and N-CMAPSS the intervals were too wide to decide; on PHM08 and FD004 the constant was significantly *better* than the model's estimate (−6.4 and −6.0 points), and on the battery cells the estimate was significantly better (+9.8 points), the one dataset where the estimate's spread at the alarm was a sizeable fraction of the window width (6.3 of 45 cycles). This

Table 3

Failure before replacement with a whole group held out (battery: lead time 30 cycles; N-CMAPSS: 10 cycles), mean of three repetitions. Pilot: CNA calibrated on m randomly chosen units of the held-out group and tested on the rest (mean over 300 draws); with $m = 5$, $\alpha = 0.10$ is not attainable and $\alpha = 0.20$ is shown.

Held-out group (units)	CNA 0.10, other groups	CNA-DA	Pilot $m = 10$, $\alpha = 0.10$	Pilot $m = 5$, $\alpha = 0.20$	STW window	STW 1-SE	Kamariotis P1 opt.	Trend 1-SE
Battery batch 1 (36)	0.00	0.00	0.095	0.167	0.00	0.00	0.00	0.00
Battery batch 2 (43)	0.74	0.45	0.094	0.165	0.35	0.00	0.12	0.00
Battery batch 3 (44)	0.61	0.36	0.084	0.167	0.06	0.02	0.17	0.00
N-CMAPSS, 6 subsets (9–15 each)	0.00–0.02	0.00	–	0.165–0.177	0.00–0.02	0.00	0.00	–
N-CMAPSS subset 4 (10)	0.33	0.10	–	0.165	0.07	0.10	0.07	–
N-CMAPSS subset 8 (15)	0.29	0.18	–	0.169	0.11	0.18	0.11	–
N-CMAPSS subset 9 (10)	0.40	0.07	–	0.169	0.10	0.13	0.03	–

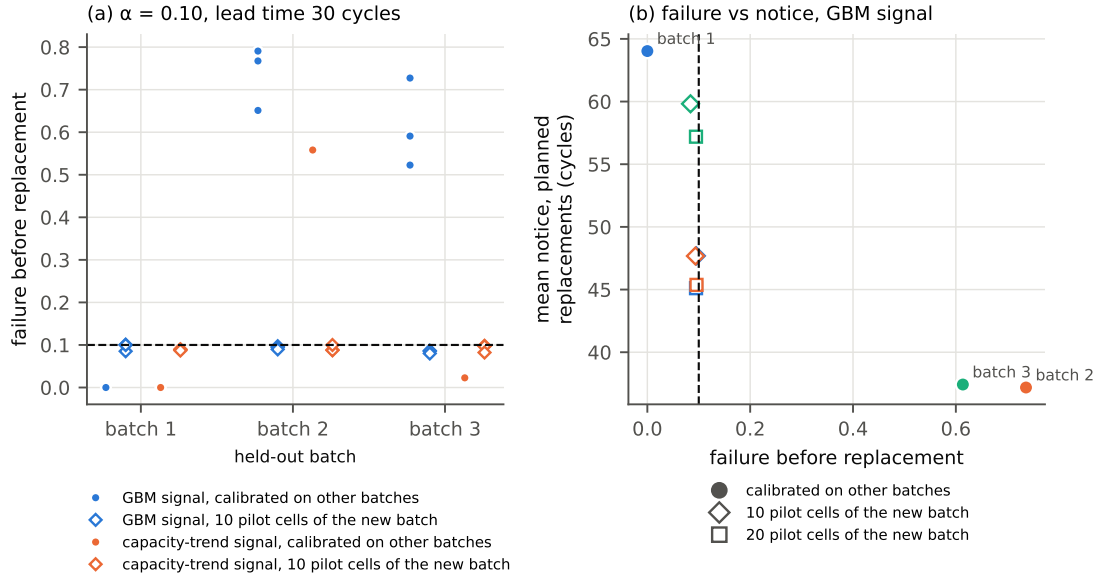


Fig. 3: Battery cells with a whole batch held out ($\alpha = 0.10$, lead time 30 cycles). (a) Failure before replacement when the alarm is calibrated on the other batches (dots: three repetitions) or on 10 pilot cells of the held-out batch (diamonds: mean over 300 random draws). (b) Failure against mean notice at planned replacements for the gradient-boosted signal.

corrects the earlier summary that the estimate changed the score by at most 3.6 points [23]. The plug-in bound of part (ii), computed on calibration units, exceeded the realised score difference on 6 of 7 datasets and the realised discordance on 4 of 7; it uses estimated probabilities and is not a guarantee. For this article the conclusion is the one that motivates the method: a window score bounds neither the notice nor the failure probability, so neither should be inferred from it.

Table 4

Window score of the window-tuned alarm's estimate and of one constant at the same alarm times ($H = 20$ cycles; battery 60), repetition 0. Difference: estimate – constant with Tango 90 % interval; equivalence margin ± 5 points. SD at the alarm in cycles. Bound: Proposition 2(ii) estimated on calibration units; Disc.: realised share of discordant units.

Dataset	n	Estimate (%)	Constant (%)	Discordant (est./const.)	Difference (90 % CI)	TOST p	SD estimate / notice	Bound / Disc. (points)
PHM08	218	87.6	94.0	4 / 18	−6.4 (−10.3, −3.1)	0.756	2.6 / 4.9	15.9 / 10.1
FD001	100	86.0	85.0	6 / 5	+1.0 (−4.9, +7.0)	0.126	3.0 / 5.3	19.1 / 11.0
FD002	260	90.4	91.9	7 / 11	−1.5 (−4.5, +1.2)	0.028	2.4 / 5.2	13.6 / 6.9
FD003	100	84.0	86.0	6 / 8	−2.0 (−8.6, +4.5)	0.216	2.9 / 5.3	13.6 / 14.0
FD004	249	79.1	85.1	6 / 21	−6.0 (−9.7, −2.7)	0.694	2.3 / 18.0	15.2 / 10.8
N-CMAPSS	99	52.5	56.6	13 / 17	−4.0 (−13.3, +5.2)	0.431	3.2 / 11.0	28.1 / 30.3
Battery	123	95.1	85.4	13 / 1	+9.8 (+5.4, +15.3)	0.964	6.3 / 12.9	8.9 / 11.4

8. Discussion

8.1. What the results establish

Calibrating the alarm rather than the interval did what Theorem 1 says it does. On seven datasets, with between 32 and 83 calibration units, the failure-before-replacement rate followed its nominal level, while a per-cycle lower bound at the same nominal level, on the same signal and trigger, failed on up to 70 % of units. The difference is not a matter of tuning: per-cycle coverage averages over cycles chosen independently of the data, whereas an alarm is read at the first cycle at which the signal crosses a level, the cycle at which an optimistic error is most likely. Restricting the per-cycle residuals to the near-failure region repaired most of the gap on six datasets, but how close it came depended on that choice of rows, and on N-CMAPSS it exceeded its level by half. Calibrating the critical level removes the choice, because the quantity calibrated is the event the operator cares about. The practical message for the growing use of conformal RUL intervals in maintenance [4–6] is that interval coverage should not be read as alarm reliability.

8.2. A deployment recipe

Table 5 turns the results into a procedure. Its steps are the ones that, in this study, changed whether the stated reliability was achieved; none requires a model of the degradation or of the RUL error.

8.3. What the guarantee costs

8.4. Transfer and censoring in the field

The battery results show the limit of every exchangeability-based guarantee: calibrated on two batches, the alarm failed on most cells of a third, shorter-lived batch, and similar failures occurred on three of nine N-CMAPSS subsets. This is not a defect of the conformal step, which did what it promised for the population it was calibrated on, but it is a warning against reading a validated failure rate as a property of the alarm rather than of the alarm and a population. Two remedies were examined. Conservative uncertified rules carried enough margin to survive this shift, at the price of notice that is wasted when the population does not change. Recalibration on ten pilot cells of the

Table 5

Deployment recipe for a notice-guaranteed end-of-life alarm.

Step and reason
1 State the lead time ℓ and the tolerated probability α of failure before the planned replacement; check $n \geq 1/\alpha - 1$ calibration units are available (Proposition 1).
2 Fix the signal and trigger (smoothing, debounce, cap) before looking at the calibration units, or select them on units not used for calibration.
3 Collect calibration units from the population in which the alarm will run. Use replaced or still-running units as censored units, their critical level computed as if they had failed when last seen alive (Theorem 2).
4 Compute critical levels (Lemma 2) and set $\hat{H}$ to their conformal quantile; with a cost ratio, choose among admissible order statistics (Corollary 1).
5 When the population changes (new batch, campaign or operating regime), recalibrate on units of the new population, starting from a conservative interim level; do not carry a calibrated level across populations.
6 Report the failure rate with its interval, the notice given, and the number of calibration units the claim rests on; do not report interval coverage or a window score as alarm reliability.

new batch restored the nominal rate, and Theorem 2 removes the need to run those cells to failure: cells replaced under a conservative interim level count as censored calibration units. Weighted or non-exchangeable conformal methods [21, 22] could use covariates of the new batch, such as its charging protocol, to reweight calibration units; this needs a model of the shift and was not attempted.

8.5. Limitations

Data. Five of the seven datasets come from one simulator family and the battery cells are laboratory data. Censoring was imposed on complete run-to-failure records by three mechanisms; no field fleet with genuinely censored lives was analysed, and none was available to us. Bearings were excluded because 17 units cannot support the guarantee.

• **Exchangeability.** The guarantee is marginal and requires exchangeable calibration and new units. It failed across battery batches and N-CMAPSS subsets, and it may fail under slower drifts within one population. It is not conditional on the calibration

set; a training-conditional version follows from the Beta distribution of the coverage [32].

- **Scope.** The theory covers threshold triggers with debounce on a causal signal and a fixed lead time. The settings of CNA (cap, smoothing, debounce) were fixed a priori, not optimised. The cross-fitted guarantee (Theorem 3) is weaker than the split guarantee.
- **Cost.** The renewal–reward model assumes independent units, replacement as good as new and illustrative cost ratios. Section ?? uses the decision setting of Kamariotis et al. but our own implementation of their policy and of the probability model behind it.
- **Pre-specification.** Both protocols were hashed before their runs but not registered, and the amendment was written after the main results were known. The equivalence and cost hypotheses of the main protocol failed and are reported as such; the prominence of Proposition 1(ii) as the explanation of the cost results was decided after those results were seen.

9. Conclusion

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to search the literature, write and run the analysis code, draft the theory and proofs, and draft and revise the manuscript text. The protocols, all experiments and all reported numbers were produced by the code in the accompanying repository and checked against its output files. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data and code availability

PHM08, C-MAPSS and N-CMAPSS are available from the NASA Prognostics Center of Excellence data repository, and the battery data of Severson et al. [25] from <https://data.matr.io>. Code, hashed protocols, derived data summaries, per-unit results and the scripts that produce every table and figure are in a repository whose link is withheld for anonymised review.

References

- [1] Saxena A, Goebel K, Simon D, Eklund N. Damage propagation modeling for aircraft engine run-to-failure simulation. In: 2008 International Conference on Prognostics and Health Management. IEEE; 2008, p. 1–9. <https://doi.org/10.1109/PHM.2008.4711414>
- [2] Saxena A, Celaya J, Saha B, Saha S, Goebel K. Metrics for offline evaluation of prognostic performance. International Journal of Prognostics and Health Management 2010;1(1). <https://doi.org/10.36001/ijphm.2010.v1i1.1336>
- [3] Kamariotis A, Tatsis K, Chatzi E, Goebel K, Straub D. A metric for assessing and optimizing data-driven prognostic algorithms for predictive maintenance. Reliability Engineering & System Safety 2024;242:109723. <https://doi.org/10.1016/j.res.2023.109723>
- [4] Javanmardi A, Hüllermeier E. Conformal prediction intervals for remaining useful lifetime estimation. International Journal of Prognostics and Health Management 2023;14(2). <https://doi.org/10.36001/ijphm.2023.v14i2.3417>
- [5] Robinson CD. Remaining useful life estimation for aircraft engines with risk-aware prediction intervals via conformalized quantile regression. International Journal of Prognostics and Health Management 2026 (published online April 2026).
- [6] Wang W, Wang Z, Cai Z, Hu C, Si S. Robust uncertainty quantification for online remaining useful life prediction with randomly missing and partially faulty sensor data. Reliability Engineering & System Safety 2025;262.
- [7] Ramasso E, Saxena A. Performance benchmarking and analysis of prognostic methods for CMAPSS datasets. International Journal of Prognostics and Health Management 2014;5(2). <https://doi.org/10.36001/ijphm.2014.v5i2.2236>
- [8] de Pater I, Reijns A, Mitici M. Alarm-based predictive maintenance scheduling for aircraft engines with imperfect remaining useful life prognostics. Reliability Engineering & System Safety 2022;221:108341. <https://doi.org/10.1016/j.res.2022.108341>
- [9] Nguyen KTP, Medjaher K. A new dynamic predictive maintenance framework using deep learning for failure prognostics. Reliability Engineering & System Safety 2019;188:251–262. <https://doi.org/10.1016/j.res.2019.03.018>
- [10] Mitici M, de Pater I, Barros A, Zeng Z. Dynamic predictive maintenance for multiple components using data-driven probabilistic RUL prognostics: The case of turbofan engines. Reliability Engineering & System Safety 2023;234:109199. <https://doi.org/10.1016/j.res.2023.109199>
- [11] Barlow R, Hunter L. Optimum preventive maintenance policies. Operations Research 1960;8(1):90–100. <https://doi.org/10.1287/opre.8.1.90>
- [12] de Jonge B, Scarf PA. A review on maintenance optimization. European Journal of Operational Research 2020;285(3):805–824. <https://doi.org/10.1016/j.ejor.2019.09.047>
- [13] Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. New York: Springer; 2005. <https://doi.org/10.1007/b106715>
- [14] Lei J, G'Sell M, Rinaldo A, Tibshirani RJ, Wasserman L. Distribution-free predictive inference for regression. Journal of the American Statistical Association 2018;113(523):1094–1111. <https://doi.org/10.1080/01621459.2017.1307116>
- [15] Angelopoulos AN, Bates S. Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning 2023;16(4):494–591. <https://doi.org/10.1561/22000000101>
- [16] Romano Y, Patterson E, Candès E. Conformalized quantile regression. In: Advances in Neural Information Processing Systems 32 (NeurIPS 2019); 2019.
- [17] Candès E, Lei L, Ren Z. Conformalized survival analysis. Journal of the Royal Statistical Society Series B 2023;85(1):24–45. <https://doi.org/10.1093/jrsssb/qkac004>
- [18] Angelopoulos AN, Bates S, Fisch A, Lei L, Schuster T. Conformal risk control. In: International Conference on Learning Representations (ICLR); 2024.
- [19] Gupta C, Kuchibhotla AK, Ramdas A. Nested conformal prediction and quantile out-of-bag ensemble methods. Pattern Recognition 2022;127:108496. <https://doi.org/10.1016/j.patcog.2021.108496>
- [20] Barber RF, Candès EJ, Ramdas A, Tibshirani RJ. Predictive inference with the jackknife+. Annals of Statistics 2024;52(1):1–25. <https://doi.org/10.1214/23-AOS2237>

- 2021;49(1):486–507. <https://doi.org/10.1214/20-AOS1965>
- [21] Tibshirani RJ, Foygel Barber R, Candès E, Ramdas A. Conformal prediction under covariate shift. In: *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*; 2019.
- [22] Barber RF, Candès EJ, Ramdas A, Tibshirani RJ. Conformal prediction beyond exchangeability. *Annals of Statistics* 2023;51(2):816–845. <https://doi.org/10.1214/23-AOS2276>
- [23] Anonymous (authors of the present study). What does an end-of-life alarm score measure? Separating prediction accuracy, alarm timing, maintenance cost and transfer. Earlier manuscript version, 2026 (superseded by this article; numbers attributed to it are prior results).
- [24] Arias Chao M, Kulkarni C, Goebel K, Fink O. Aircraft engine run-to-failure dataset under real flight conditions for prognostics and diagnostics. *Data* 2021;6(1):5. <https://doi.org/10.3390/data6010005>
- [25] Severson KA, Attia PM, Jin N, Perkins N, Jiang B, Yang Z, Chen MH, Aykol M, Herring PK, Fraggedakis D, Bazant MZ, Harris SJ, Chueh WC, Braatz RD. Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy* 2019;4(5):383–391. <https://doi.org/10.1038/s41560-019-0356-8>
- [26] Nectoux P, Gouriveau R, Medjaher K, Ramasso E, Chebel-Morello B, Zerhouni N, Varnier C. PRONOSTIA: An experimental platform for bearings accelerated degradation tests. In: *IEEE International Conference on Prognostics and Health Management (PHM'12)*, Denver, CO; 2012. <https://hal.science/hal-00719503>
- [27] Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 2011;12:2825–2830.
- [28] Clopper CJ, Pearson ES. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika* 1934;26(4):404–413. <https://doi.org/10.1093/biomet/26.4.404>
- [29] Holm S. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 1979;6(2):65–70.
- [30] Tango T. Equivalence test and confidence interval for the difference in proportions for the paired-sample design. *Statistics in Medicine* 1998;17(8):891–908.
- [31] Schuirman DJ. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics* 1987;15(6):657–680. <https://doi.org/10.1007/BF01068419>
- [32] Vovk V. Conditional validity of inductive conformal predictors. In: *Proceedings of the Asian Conference on Machine Learning*, PMLR 25; 2012, p. 475–490.